Fiscal Year:	FY 2005	**Task Last Updated:**	FY 09/05/2006
PI Name:	Lackner, James R. Ph.D.		
Project Title:	Predicting and Assessing Postural Reentry Disturbances		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02454-9110	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	09/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2008 per P. Denkins/JSC (5/07) NOTE: Funding reinstated to 6/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bortolami , Simone (Brandeis University) DiZio, Paul (Brandeis University)		
Grant/Contract No.:	NNJ04HJ14G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We have developed a paradigm for abruptly perturbing posture which produces transient sway that can be closely fit by a four link inverted pendulum model. Preliminary evidence shows that four fitted parameters of the model quantify the loss of postural control caused by a simulated change in gravitoinertial force background. We will use our model and paradigm to quantify the destabilizing effects of and the recovery from actual exposure to gravitoinertial force transitions, in parabolic flight, in a rotating artificial gravity environment, and following body loading and unloading in 1g. This research is directly relevant to all questions listed as pertinent to the risk entitled “Disorientation and inability to perform landing, egress, or other physical tasks, especially during/after G-level changes” (Risk No. 33) in the NASA Critical Path Roadmap. This project will result in a Countermeasure Readiness Level of 5-6. Aim 1. Measure postural responses to Hold & Release in parabolic flight during conditions of weightlessness with the aid of a hold-down device and during the high force levels. Hypothesis: We predict that body segment dynamics and EMG patterns will be modulated by the gravity force level. Aim 2. Measure postural responses to Hold & Release before and after adaptation to artificially increased or decreased body weight. Hypothesis: We predict that adapted postural behavior will show parallel changes to those observed under altered force levels in parabolic flight. Aim 3. Measure postural responses to Hold & Release before, during, and after exposure to increased gravitoinertial force level in a rotating room. Hypothesis: Postural behavior will show decreased characteristic frequencies per-rotation and post-rotation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This is the first year of this three year project, and we have completed Aim 1. A manuscript is near ready for submission. We elicited loss of balance using the Hold and Release (H&R) paradigm onboard NASA’s KC-135 aircraft. The aircraft produces 25s long steady-state 0 and 1.8g lapses of time by flying parabolic trajectories. We conducted trials during the 0 and 1.8g phases of the flight, and in 1g level flight (control) both with and without the aid of vision. 0g trials were conducted with 50% body weight bungee cord loading. We also tested subjects pre- and post-flight. A horizontal force was applied to the subject’s sternum, which he or she actively resisted. Removing the force caused subjects to sway forward and they had to react to prevent falling. We measured EMG onset latency in each muscle after release. We are currently performing analyses to fit the stiffness and damping parameters of the inverted pendulum model to these data. We expect to find transient post-flight changes relative to pre-flight baselines with gradual recovery. Relationship to NASA critical path: This work is most related to the Neurovestibular category of the critical path roadmap. The negative outcome of spaceflight that we are addressing is the demonstrated postural and locomotory instability that occur after Shuttle missions or longer duration habitation in space. This is a consequence of gross alterations in vestibular and neuromuscular function (risk factors) which are caused by exposure to microgravity (initiating event). This work will quantify the recovery of postural inegity post-flight and will provide a means of in-flight testing of loss of postural integrity and of the effectiveness of countermeasures designed to prevent this loss.
Bibliography Type:	Description: (Last Updated: 03/31/2017)
Articles in Peer-reviewed Journals	Hudson TE, Lackner JR, DiZio P. "Rapid adaptation of torso pointing movements to perturbations of the base of support." Exp Brain Res., in press 2005. , Jul-2005
Articles in Peer-reviewed Journals	Kurtzer I, DiZio PA, Lackner JR. "Adaptation to a novel multi-force environment." Exp Brain Res. 2005 Jul;164(1):120-32. Epub 2005 Apr 16. PMID: 15834711 , Jul-2005
Books/Book Chapters	Lackner JR, DiZio P. "Multisensory influences on orientation and movement control." in "The handbook of multisensory processes." Ed. Gemma A. Calvert, Charles Spence, Barry E. Stein. Cambridge, Mass. : MIT Press, c2004., Aug-2004
Books/Book Chapters	Lackner JR, DiZio P. "Cyber Adaptation Syndrome." in "Encyclopedia of Neuroscience, 3rd edition, CD-ROM version." Ed. George Adelman, Barry H. Smith. Amsterdam, Netherlands : Elsevier Science BV, c2004., Sep-2004